9.8 Line Integrals

Introduction

The notion of the definite integral $\int_a^b f(x)\,dx$; that is, *integration of a function defined over an interval*, can be generalized to *integration of a function defined along a curve*. To this end we need to introduce some terminology about curves.

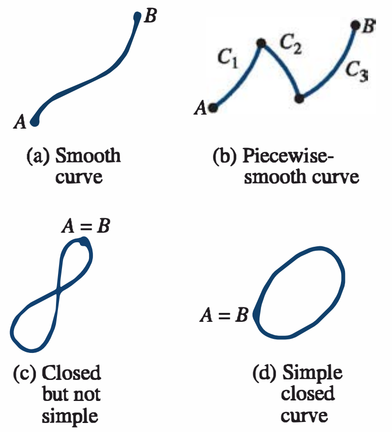


FIGURE 9.8.1 Various curves

Terminology

Suppose C is a curve parameterized by $x = f(t)$, $y = g(t)$, $a \le t \le b$, and A and B are the points $(f(a), g(a))$ and $(f(b), g(b))$, respectively. We say that

(*i*) C is a **smooth curve** if f' and g' are continuous on the closed interval $[a, b]$ and not simultaneously zero on the open interval (a, b).

(*ii*) C is **piecewise smooth** if it consists of a finite number of smooth curves $C_1, C_2, \ldots, C_n$ joined end to end—that is, $C = C_1 \cup C_2 \cup \cdots \cup C_n$.

(*iii*) C is a **closed curve** if $A = B$.

(*iv*) C is a **simple closed curve** if $A = B$ and the curve does not cross itself.

(*v*) If C is not a closed curve, then the **positive direction** on C is the direction corresponding to increasing values of t.

FIGURE 9.8.1 illustrates each type of curve defined in (*i*)–(*iv*).

This same terminology carries over in a natural manner to curves in space. For example, a curve C defined by $x = f(t)$, $y = g(t)$, $z = h(t)$, $a \le t \le b$, is smooth if f', g', and h' are continuous on $[a, b]$ and not simultaneously zero on (a, b).

Definite Integral

Before defining integration along a curve, let us review the five steps leading to the definition of the definite integral.

1. Let $y = f(x)$ be defined on a closed interval $[a, b]$.
2. Partition the interval $[a, b]$ into n subintervals $[x_{k-1}, x_k]$ of lengths $\Delta x_k = x_k - x_{k-1}$. Let P denote the partition shown in FIGURE 9.8.2(a).
3. Let $\|P\|$ be the length of the longest subinterval. The number $\|P\|$ is called the **norm** of the partition P.
4. Choose a sample point x_k^* in each subinterval. See Figure 9.8.2(b).
5. Form the sum $\displaystyle\sum_{k=1}^{n} f(x_k^*)\Delta x_k$.

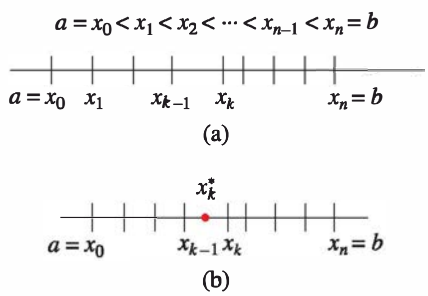


FIGURE 9.8.2 Sample point in kth subinterval

The definite integral of a function of a single variable is given by the limit of a sum:

$$\int_a^b f(x)\,dx = \lim_{\|P\|\to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k.$$

Line Integrals in the Plane

The following analogous five steps lead to the definitions of three **line integrals*** in the plane.

1. Let $z = G(x, y)$ be defined in some region that contains the smooth curve C defined by $x = f(t)$, $y = g(t)$, $a \le t \le b$.
2. Divide C into n subarcs of lengths Δs_k according to the partition $a = t_0 < t_1 < t_2 < \cdots < t_n = b$ of $[a, b]$. Let the projection of each subarc onto the x- and y-axes have lengths Δx_k and Δy_k, respectively.
3. Let $\|P\|$ be the **norm** of the partition or the length of the longest subarc.
4. Choose a sample point (x_k^*, y_k^*) on each subarc. See FIGURE 9.8.3.
5. Form the sums

$$\sum_{k=1}^{n} G(x_k^*, y_k^*)\,\Delta x_k,\quad \sum_{k=1}^{n} G(x_k^*, y_k^*)\,\Delta y_k,\quad \sum_{k=1}^{n} G(x_k^*, y_k^*)\,\Delta s_k.$$

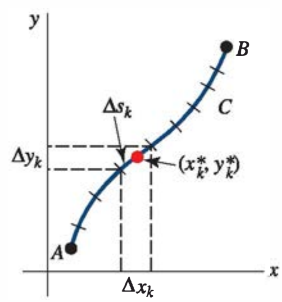


FIGURE 9.8.3 Sample point in kth subarc

*An unfortunate choice of names. **Curve integrals** would be more appropriate.

Definition 9.8.1 Line Integrals in the Plane

Let G be a function of two variables x and y defined on a region of the plane containing a smooth curve C.

(*i*) The **line integral of G along C from A to B with respect to x** is

$$\int_C G(x, y)\, dx = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\, \Delta x_k.$$

(*ii*) The **line integral of G along C from A to B with respect to y** is

$$\int_C G(x, y)\, dy = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\, \Delta y_k.$$

(*iii*) The **line integral of G along C from A to B with respect to arc length** is

$$\int_C G(x, y)\, ds = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\, \Delta s_k.$$

It can be proved that if $G(x, y)$ is continuous on C, then the integrals defined in (*i*), (*ii*), and (*iii*) exist. We shall assume continuity of G as a matter of course.

Method of Evaluation—Curve Defined Parametrically The line integrals in Definition 9.8.1 can be evaluated in two ways, depending on whether the curve C is defined parametrically or by an explicit function. In either case the basic idea is to convert the line integral to a definite integral in a single variable. If C is a smooth curve parameterized by $x = f(t)$, $y = g(t)$, $a \le t \le b$, then we simply replace x and y in the integral by the functions $f(t)$ and $g(t)$, and the appropriate differential dx, dy, or ds by $f'(t)\, dt$, $g'(t)\, dt$, or $\sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt$. The expression $ds = \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt$ is called the **differential of arc length**. The integration is carried out with respect to the variable t in the usual manner:

$$\int_C G(x, y)\, dx = \int_a^b G(f(t), g(t))\, f'(t)\, dt, \tag{1}$$

$$\int_C G(x, y)\, dy = \int_a^b G(f(t), g(t))\, g'(t)\, dt, \tag{2}$$

$$\int_C G(x, y)\, ds = \int_a^b G(f(t), g(t))\sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt. \tag{3}$$

EXAMPLE 1 Evaluation of Line Integrals

Evaluate **(a)** $\int_C xy^2\, dx$, **(b)** $\int_C xy^2\, dy$, and **(c)** $\int_C xy^2\, ds$ on the quarter-circle C defined by $x = 4\cos t$, $y = 4\sin t$, $0 \le t \le \pi/2$. See **FIGURE 9.8.4**.

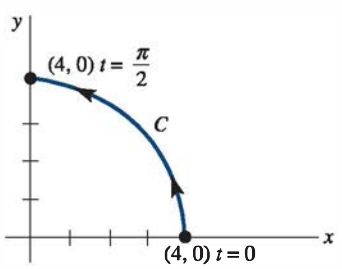


FIGURE 9.8.4 Curve C in Example 1

SOLUTION **(a)** From (1),

$$\int_C xy^2\, dx = \int_0^{\pi/2} \overbrace{(4\cos t)}^{x}\overbrace{(16\sin^2 t)}^{y^2}\overbrace{(-4\sin t\, dt)}^{dx}$$

$$= -256\int_0^{\pi/2} \sin^3 t\cos t\, dt$$

$$= -256\left[\frac{1}{4}\sin^4 t\right]_0^{\pi/2} = -64.$$

(b) From (2),

$$\int_C xy^2\,dy = \int_0^{\pi/2} \overbrace{(4\cos t)}^{x}\overbrace{(16\sin^2 t)}^{y^2}\overbrace{(4\cos t\,dt)}^{dy}$$

$$= 256\int_0^{\pi/2} \sin^2 t\cos^2 t\,dt$$

$$= 256\int_0^{\pi/2} \frac{1}{4}\sin^2 2t\,\,dt \qquad \leftarrow \text{trigonometric identities}$$

$$= 64\int_0^{\pi/2} \frac{1}{2}(1 - \cos 4t)\,dt$$

$$= 32\left[t - \frac{1}{4}\sin 4t\right]_0^{\pi/2} = 16\pi.$$

(c) From (3),

$$\int_C xy^2\,ds = \int_0^{\pi/2} \overbrace{(4\cos t)}^{x}\overbrace{(16\sin^2 t)}^{y^2}\overbrace{\sqrt{16(\cos^2 t + \sin^2 t)}\,dt}^{ds}$$

$$= 256\int_0^{\pi/2} \sin^2 t\cos t\,dt$$

$$= 256\left[\frac{1}{3}\sin^3 t\right]_0^{\pi/2} = \frac{256}{3}.$$

≡

Method of Evaluation—Curve Defined by an Explicit Function If the curve C is defined by an explicit function $y = f(x)$, $a \le x \le b$, we can use x as a parameter. With $dy = f'(x)\,dx$ and $ds = \sqrt{1 + [f'(x)]^2}\,dx$, the foregoing line integrals become, in turn,

$$\int_C G(x, y)\,dx = \int_a^b G(x, f(x))\,dx, \tag{4}$$

$$\int_C G(x, y)\,dy = \int_a^b G(x, f(x))f'(x)\,dx, \tag{5}$$

$$\int_C G(x, y)\,ds = \int_a^b G(x, f(x))\sqrt{1 + [f'(x)]^2}\,dx. \tag{6}$$

A line integral along a *piecewise-smooth* curve C is defined as the *sum* of the integrals over the various smooth curves whose union comprises C. For example, if C is composed of smooth curves C_1 and C_2, then

$$\int_C G(x, y)\,ds = \int_{C_1} G(x, y)\,ds + \int_{C_2} G(x, y)\,ds.$$

Notation In many applications, line integrals appear as a sum

$$\int_C P(x, y)\,dx + \int_C Q(x, y)\,dy.$$

It is common practice to write this sum as one integral without parentheses as

$$\int_C P(x, y)\,dx + Q(x, y)\,dy \quad \text{or simply} \quad \int_C P\,dx + Q\,dy. \tag{7}$$

A line integral along a *closed* curve C is very often denoted by

$$\oint_C P\,dx + Q\,dy.$$

EXAMPLE 2 Curve Defined by an Explicit Function

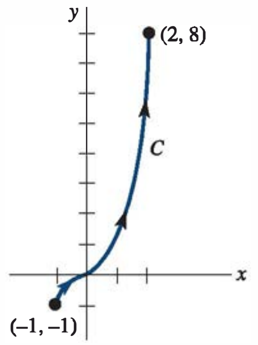


FIGURE 9.8.5 Curve C in Example 2

Evaluate $\int_C xy\,dx + x^2\,dy$, where C is given by $y = x^3$, $-1 \le x \le 2$.

SOLUTION The curve C is illustrated in **FIGURE 9.8.5** and is defined by the explicit function $y = x^3$. Hence we can use x as the parameter. Using $dy = 3x^2\,dx$, we get

$$\begin{aligned}\int_C xy\,dx + x^2\,dy &= \int_{-1}^{2} x\overbrace{(x^3)}^{y}\,dx + x^2\overbrace{(3x^2\,dx)}^{dy}\\ &= \int_{-1}^{2} 4x^4\,dx\\ &= \frac{4}{5}x^5\Big]_{-1}^{2} = \frac{132}{5}.\end{aligned}$$

≡

EXAMPLE 3 Curve Defined Parametrically

Evaluate $\oint_C x\,dx$, where C is the circle $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$.

SOLUTION From (1),

$$\oint_C x\,dx = \int_0^{2\pi} \cos t\,(-\sin t\,dt) = \frac{1}{2}\cos^2 t\Big]_0^{2\pi} = \frac{1}{2}[1 - 1] = 0.$$

≡

EXAMPLE 4 Closed Curve

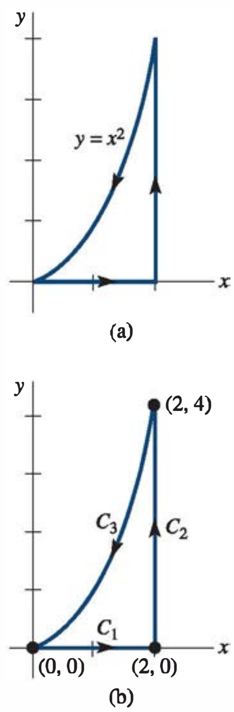


FIGURE 9.8.6 Curve C in Example 4

Evaluate $\oint_C y^2\,dx - x^2\,dy$ on the closed curve C that is shown in **FIGURE 9.8.6(a)**.

SOLUTION Since C is piecewise smooth, we express the integral as a sum of integrals. Symbolically, we write

$$\oint_C = \int_{C_1} + \int_{C_2} + \int_{C_3}$$

where C_1, C_2, and C_3 are the curves shown in Figure 9.8.6(b). On C_1, we use x as a parameter. Since $y = 0$, $dy = 0$; therefore,

$$\int_{C_1} y^2\,dx - x^2\,dy = \int_0^2 0\,dx - x^2(0) = 0.$$

On C_2, we use y as a parameter. From $x = 2$, $dx = 0$, we have

$$\begin{aligned}\int_{C_2} y^2\,dx - x^2\,dy &= \int_0^4 y^2(0) - 4\,dy\\ &= -\int_0^4 4\,dy = -4y\Big]_0^4 = -16.\end{aligned}$$

Finally, on C_3, we again use x as a parameter. From $y = x^2$, we get $dy = 2x\,dx$ and so

$$\int_{C_3} y^2\,dx - x^2\,dy = \int_2^0 x^4\,dx - x^2(2x\,dx)$$

$$= \int_2^0 (x^4 - 2x^3)\,dx$$

$$= \left(\frac{1}{5}x^5 - \frac{1}{2}x^4\right)\Big]_2^0 = \frac{8}{5}.$$

Hence, $$\int_C y^2\,dx - x^2\,dy = 0 - 16 + \frac{8}{5} = -\frac{72}{5}.$$ ≡

It is important to be aware that a line integral is independent of the parameterization of the curve C provided C is given the same orientation by all sets of parametric equations defining the curve. See Problem 37 in Exercises 9.8. Also, recall for definite integrals that $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$. Line integrals possess a similar property. Suppose, as shown in **FIGURE 9.8.7**, that $-C$ denotes the curve having the opposite orientation of C. Then it can be shown that

$$\int_{-C} P\,dx + Q\,dy = -\int_C P\,dx + Q\,dy,$$

or equivalently,

$$\int_{-C} P\,dx + Q\,dy + \int_C P\,dx + Q\,dy = 0. \tag{8}$$

For example, in part (*a*) of Example 1, $\int_{-C} xy^2\,dx = 64$.

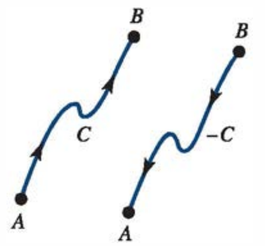


FIGURE 9.8.7 Curves with opposite orientation

Line Integrals in Space

Line integrals of a function G of three variables, $\int_C G(x, y, z)\,dx$, $\int_C G(x, y, z)\,dy$, and $\int_C G(x, y, z)\,ds$, are defined in a manner analogous to Definition 9.8.1. However, to that list we add a fourth **line integral along a space curve C with respect to z:**

$$\int_C G(x, y, z)\,dz = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*, z_k^*)\,\Delta z_k. \tag{9}$$

Method of Evaluation

If C is a smooth curve in 3-space defined by the parametric equations $x = f(t)$, $y = g(t)$, $z = h(t)$, $a \le t \le b$, then the integral in (9) can be evaluated by using

$$\int_C G(x, y, z)\,dz = \int_a^b G(f(t), g(t), h(t))\,h'(t)\,dt.$$

The integrals $\int_C G(x, y, z)\,dx$ and $\int_C G(x, y, z)\,dy$ are evaluated in a similar fashion. The line integral with respect to arc length is

$$\int_C G(x, y, z)\,ds = \int_a^b G(f(t), g(t), h(t))\sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}\,dt.$$

As in (7), in 3-space we are often concerned with line integrals in the form of a sum:

$$\int_C P(x, y, z)\,dx + Q(x, y, z)\,dy + R(x, y, z)\,dz.$$

EXAMPLE 5 Line Integral on a Curve in 3-Space

Evaluate $\int_C y\,dx + x\,dy + z\,dz$, where C is the helix $x = 2\cos t$, $y = 2\sin t$, $z = t$, $0 \le t \le 2\pi$.

SOLUTION Substituting the expressions for x, y, and z along with $dx = -2\sin t\,dt$, $dy = 2\cos t\,dt$, $dz = dt$, we get

$$\int_C y\,dx + x\,dy + z\,dz = \int_0^{2\pi} (\underbrace{-4\sin^2 t + 4\cos^2 t}_{\text{double angle formula}})\,dt + t\,dt$$

$$= \int_0^{2\pi} (4\cos 2t + t)\,dt$$

$$= \left(2\sin 2t + \frac{t^2}{2}\right)\Big]_0^{2\pi} = 2\pi^2.$$ ≡

We can use the concept of a vector function of several variables to write a general line integral in a compact fashion. For example, suppose the vector-valued function $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is defined along a curve C: $x = f(t)$, $y = g(t)$, $a \le t \le b$, and suppose $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$ is the position vector of points on C. Then the derivative of $\mathbf{r}(t)$,

$$\frac{d\mathbf{r}}{dt} = f'(t)\mathbf{i} + g'(t)\mathbf{j} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j},$$

prompts us to define $d\mathbf{r} = \dfrac{d\mathbf{r}}{dt}\,dt = dx\mathbf{i} + dy\mathbf{j}$. Since $\mathbf{F}(x, y) \cdot d\mathbf{r} = P(x, y)\,dx + Q(x, y)\,dy$ we can write

$$\int_C P(x, y)\,dx + Q(x, y)\,dy = \int_C \mathbf{F} \cdot d\mathbf{r}. \tag{10}$$

Similarly, for a line integral on a space curve,

$$\int_C P(x, y, z)\,dx + Q(x, y, z)\,dy + R(x, y, z)\,dz = \int_C \mathbf{F} \cdot d\mathbf{r}, \tag{11}$$

where $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ and $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$.

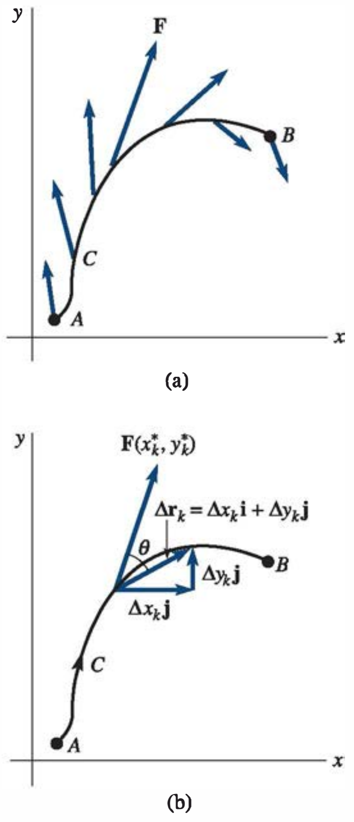


FIGURE 9.8.8 Force field $\mathbf{F}$ varies along curve C

Work In Section 7.3 we saw that the work W done by a constant force $\mathbf{F}$ that causes a straight-line displacement $\mathbf{d}$ of an object is $W = \mathbf{F} \cdot \mathbf{d}$. In beginning courses in calculus or physics it is then shown that the work done in moving an object from $x = a$ to $x = b$ by a force $F(x)$, which varies in magnitude but not in direction, is given by the definite integral $W = \int_a^b F(x)\,dx$. In general, a force field $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ acting at each point on a smooth curve C: $x = f(t)$, $y = g(t)$, $a \le t \le b$, varies in both magnitude and direction. See **FIGURE 9.8.8(a)**. If A and B are the points $(f(a), g(a))$ and $(f(b), g(b))$, respectively, we ask: What is the work done by $\mathbf{F}$ as its point of application moves along C from A to B? To answer this question, suppose C is divided into n subarcs of lengths Δs_k. On each subarc $\mathbf{F}(x_k^*, y_k^*)$ is a constant force. If, as shown in Figure 9.8.8(b), the length of the vector $\Delta\mathbf{r}_k = (x_k - x_{k-1})\mathbf{i} + (y_k - y_{k-1})\mathbf{j} = \Delta x_k\mathbf{i} + \Delta y_k\mathbf{j}$ is an approximation to the length of the kth subarc, then the approximate work done by $\mathbf{F}$ over the subarc is

$$\begin{aligned}(\|\mathbf{F}(x_k^*, y_k^*)\| \cos\theta)\|\Delta\mathbf{r}_k\| &= \mathbf{F}(x_k^*, y_k^*) \cdot \Delta\mathbf{r}_k \\ &= P(x_k^*, y_k^*)\,\Delta x_k + Q(x_k^*, y_k^*)\,\Delta y_k.\end{aligned}$$

By summing these elements of work and passing to the limit, we naturally define the **work done by F along C** as the line integral

$$W = \int_C P(x, y)\,dx + Q(x, y)\,dy \quad \text{or} \quad W = \int_C \mathbf{F} \cdot d\mathbf{r}. \tag{12}$$

Of course, (12) extends to force fields acting at points on a space curve. In this case, work $\int_C \mathbf{F} \cdot d\mathbf{r}$ is defined as in (11).

Now, since

$$\frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt},$$

we let $d\mathbf{r} = \mathbf{T}\,ds$, where $\mathbf{T} = d\mathbf{r}/ds$ is a unit tangent to C. Hence,

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\,ds = \int_C \text{comp}_{\mathbf{T}}\mathbf{F}\,ds. \tag{13}$$

In other words:

The work done by a force **F** *along a curve C is due entirely to the tangential component of* **F**.

EXAMPLE 6 Work Done by a Force

Find the work done by **(a)** $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ and **(b)** $\mathbf{F} = \frac{3}{4}\mathbf{i} + \frac{1}{2}\mathbf{j}$ along the curve C traced by $\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j}$ from $t = 0$ to $t = \pi$.

SOLUTION **(a)** The vector function $\mathbf{r}(t)$ gives the parametric equations $x = \cos t$, $y = \sin t$, $0 \le t \le \pi$, which we recognize as a half-circle. As seen in **FIGURE 9.8.9**, the force field $\mathbf{F}$ is perpendicular to C at every point. Since the tangential components of $\mathbf{F}$ are zero, the work done along C is zero. To see this we use (12):

$$W = \int_C \mathbf{F}\cdot d\mathbf{r} = \int_C (x\mathbf{i} + y\mathbf{j})\cdot d\mathbf{r}$$

$$= \int_0^{\pi} (\cos t\,\mathbf{i} + \sin t\,\mathbf{j})\cdot(-\sin t\,\mathbf{i} + \cos t\,\mathbf{j})\,dt$$

$$= \int_0^{\pi} (-\cos t\sin t + \sin t\cos t)\,dt = 0.$$

(b) In **FIGURE 9.8.10** the vectors in red are the projections of $\mathbf{F}$ on the unit tangent vectors. The work done by $\mathbf{F}$ is

$$W = \int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \left(\frac{3}{4}\mathbf{i} + \frac{1}{2}\mathbf{j}\right)\cdot d\mathbf{r}$$

$$= \int_0^{\pi}\left(\frac{3}{4}\mathbf{i} + \frac{1}{2}\mathbf{j}\right)\cdot(-\sin t\,\mathbf{i} + \cos t\,\mathbf{j})\,dt$$

$$= \int_0^{\pi}\left(-\frac{3}{4}\sin t + \frac{1}{2}\cos t\right)dt$$

$$= \left(\frac{3}{4}\cos t + \frac{1}{2}\sin t\right)\Big]_0^{\pi} = -\frac{3}{2}.$$

The units of work depend on the units of $\|\mathbf{F}\|$ and on the units of distance. ≡

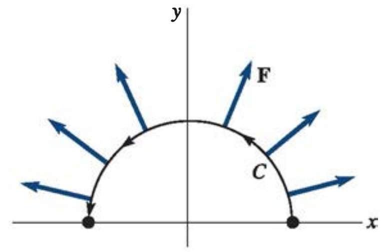


FIGURE 9.8.9 Force field in (a) of Example 6

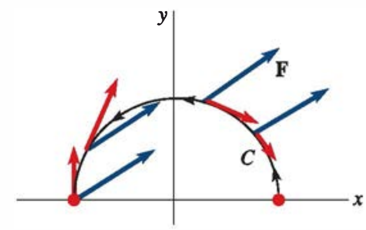


FIGURE 9.8.10 Force field in (b) of Example 6

Circulation

A line integral of a vector field $\mathbf{F}$ around a simple closed curve C is said to be the **circulation** of $\mathbf{F}$ around C; that is,

$$\text{circulation} = \oint_C \mathbf{F}\cdot d\mathbf{r} = \oint_C \mathbf{F}\cdot\mathbf{T}\,ds.$$

In particular, if $\mathbf{F}$ is the velocity field of a fluid, then the circulation is a measure of the amount by which the fluid tends to turn the curve C by rotating, or circulating, around it. For example, if $\mathbf{F}$ is perpendicular to $\mathbf{T}$ for every (x, y) on C, then $\int_C \mathbf{F}\cdot\mathbf{T}\,ds = 0$, and the curve does not move at all. On the other hand, $\int_C \mathbf{F}\cdot\mathbf{T}\,ds > 0$ and $\int_C \mathbf{F}\cdot\mathbf{T}\,ds < 0$ mean that the fluid tends to rotate C in the counterclockwise and clockwise directions, respectively. See **FIGURE 9.8.11**.

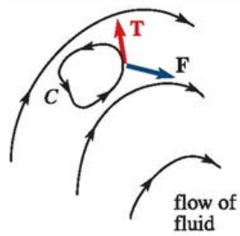
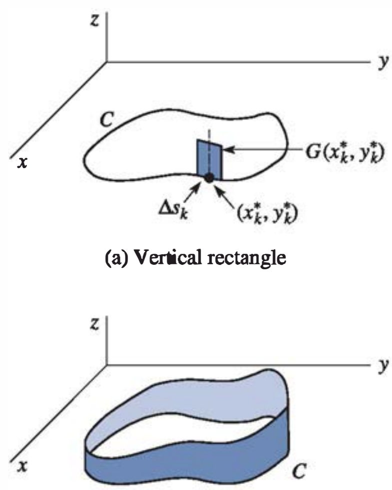


FIGURE 9.8.11 Does the velocity field turn the curve C?

Remarks

In the case of two variables, the line integral with respect to arc length $\int_C G(x, y)\,ds$ can be interpreted in a geometric manner when $G(x, y) \ge 0$ on C. In Definition 9.8.1 the symbol Δs_k represents the length of the kth subarc on the curve C. But from Figure 9.8.3 accompanying that definition, we have the approximation $\Delta s_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$. With this interpretation of Δs_k we see from **FIGURE 9.8.12(a)** that the product $G(x_k^*, y_k^*)\,\Delta s_k$ is the area of a vertical rectangle of height $G(x_k^*, y_k^*)$ and width Δs_k. The integral $\int_C G(x, y)\,ds$ then represents the area of one side of a "fence" or "curtain" extending from the curve C in the xy-plane up to the graph of $G(x, y)$ that corresponds to points (x, y) on C. See Figure 9.8.12(b).



(a) Vertical rectangle

(b) "Fence" or "curtain" of varying height $G(x, y)$ with base C

FIGURE 9.8.12 A geometric interpretation of a line integral